

Roshan Hegde

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EDUCATION

University of Illinois at Urbana-Champaign

May 2027

Bachelor of Science in Computer Science, Grainger College of Engineering

GPA: 3.80/4.0

Bachelor of Science in Liberal Arts and Sciences in Mathematics

Courses: Advanced Algorithms, Operating Systems, Programming Languages and Compilers, Abstract Algebra, Graph Theory, Cryptography, Data Structures

EXPERIENCE

Grainger | *Software Engineer Intern* | *Typescript, PostgreSQL, AWS*

June - August 2026

- Built a code migration tool used across 12 repositories to port Mocha to Vitest, cutting CI/CD times by 64%
- Paired deterministic code transformations with agentic editing to migrate frontend testing frameworks without compromising correctness, eliminating roughly 81% of manual migration work

OFI | *SWE Intern* | *Azure, Python*

May - August 2025

- Analyzed cloud asset utilization, cutting cloud spend by \$30,000 annually
- Built a self-directed data analysis tool to automate VM performance reporting and surface over-allocated resources
- Proposed a plan to improve resource cost-effectiveness to the infrastructure team

University Of Illinois Urbana-Champaign | *OS Course Assistant* | *C, RISC-V, GDB*

August 2025 - Present

- Developed programming assignments and staff solutions, and rearchitected autograder infrastructure to grade student code concurrently, improving throughput by 600%
- Hosted office hours for 1000+ students, guiding diverse valid implementations of UART drivers, thread schedulers, virtual memory, filesystem drivers, and process isolation

PROJECTS

MechMania | *Lead Developer* | *Go, Rust, Python, Nix*

August 2024 - Present

- Reworked a web-socket IPC protocol to shared memory, resulting in a >1000x speedup over the prior websocket-based IPC
- Hardened security by using Nix and bubblewrap to sandbox execution of arbitrary code

Operating System (LumonOS) | *C, Unix, RISC-V, QEMU, GDB*

January - August 2025

- Created fully functional, RISC-V, Unix-based pre-emptive operating system with round robin scheduling policy
- Built a virtio block driver, cryptography driver, and entropy driver
- Implemented support for pipes between processes and argument passing according to POSIX standard
- Placed 3rd in the Operating System design competition hosted by the ECE department

GPT2 Inference Optimization Competition | *Cuda, C, PTX, Nsight Compute*

December 2025

- Created custom CUDA kernels, profiled with NVIDIA Nsight Compute, achieving a 120x speedup over the baseline and 2x speedup over NVIDIA cuBLAS
- Implemented and tuned optimizations including flash attention, tensor cores, shared memory tiling, and fusion

AWARDS

1st Place, GPT2 Inference Optimization Competition

December 2025

1st Place, Hackillinois '24, '25 | *Rust, Cargo*

February 2024 and 2025

1st Place, MechMania '24, '25 | *Java, Python*

September 2023 and 2024

SKILLS

Languages: C, C++, Rust, Go, Java, Kotlin, Python, Typescript

Tools: Git, Docker, Linux, Nix, Vim, AWS, Azure, PostgreSQL, GDB